

1 Lecture 1

1.1 Exercise: Form groups

Go to the plan for the first lecture and register yourself to the group you wish to be part of.

2 Lecture 2

2.1 Exercise 1: Setting up your environment

Presentation from instructors about GitHub template, GitHub Actions, and general recommendations.

Deliverables:

- Each group has a GitHub repository with all group members as collaborators
- Your pipeline should run successfully, and generate the artifacts needed for the final hand-in

2.2 Exercise 2: Define Use Cases

Objective:

The goal of this exercise is to develop a deeper understanding of the case presented in the exam PowerPoint (see Lecture 1). You are to identify use cases and make assumptions based on the given information.

Tasks:

- *Examine the slides:* Based on the case study and requirements presented in the exam PowerPoint, list details that might be relevant when defining use cases.
- *List Assumptions:* Identify and enumerate any assumptions you make during your analysis. These could include:
 - Company details like size, products, departments, etc.
 - Scale, such as the number of sensors, actuators, and production facilities involved.
 - Challenges that the system may face.
- *Other Considerations:* Feel free to list any other considerations or variables that may affect the software architecture or use cases.
- *Define use cases:* Create a list of use cases for your system. It is better to list few well thought-out use cases that implies architecture relevant considerations than a long list of trivial use cases.

Example Use Case Format:

Use Case Name: Schedule Production Run *Actors:* Production Manager, Production Scheduler

Preconditions: A verified production order exists. *Steps:*

1. Production Manager logs into the production software.
2. Production Manager selects the "Schedule Production Run" option.
3. Production Scheduler fetches available time slots and resources.

4. Production Manager selects appropriate time slot and resources.
5. Production Scheduler confirms the schedule and notifies related components.

Postconditions: A production run is scheduled, and all affected components are notified.

Deliverables:

- A document outlining all the identified use cases.
- A list of assumptions made, backed with justifications.

Assessment Criteria:

- Completeness and depth of use cases identified.
- Rationality behind listed assumptions.